

AI Medical Coding

Semantic Similarity-Based ICD-10-CM Code Mapping

Objective

This experiment aims to calculate the semantic similarity between clinical documents and the descriptions of ICD-10-CM codes. The goal is to assign the most relevant ICD-10-CM code(s) to a given clinical text by leveraging pre-trained transformer-based models for semantic similarity. The hierarchical structure of ICD-10-CM codes is exploited to progressively narrow down potential matches, starting from broader categories and moving to more specific subcategories.

Methodology

Data Preparation

The inputs for this process are twofold:

1. **Clinical Text:** Free-form medical descriptions or notes containing diagnoses, symptoms, or treatment information.
2. **ICD-10-CM Dataset:** A structured set of ICD-10-CM codes organized hierarchically:
 - The first character of each code (e.g., "A", "B", "C") represents the top-level category.
 - Descriptions for each level (top-level, subcategories, and specific codes) provide a semantic basis for matching.

The ICD-10-CM hierarchy is a key component of this approach, as it allows for systematic exploration and filtering of potential matches.

Stepwise Hierarchical Matching

This approach uses a hierarchical narrowing process to find the most relevant ICD-10-CM code.

1. Top-Level Matching

The first step involves analyzing the **broad categories** of ICD-10-CM codes. The clinical text is compared with the descriptions of top-level categories, such as "A" for infectious diseases or "E" for endocrine disorders. Semantic similarity is computed, and the top-level category with the highest similarity score is selected for further exploration.

This step significantly reduces the search space, ensuring that only relevant categories are considered in subsequent stages.

2. Subcategory Matching

Once the top-level category is identified, the process moves to the next level of granularity: subcategories within that top-level category. For example, if "E" is selected, subcategories like "E10" (Type 1 diabetes mellitus) and "E11" (Type 2 diabetes mellitus) are considered. The clinical text is matched against the descriptions of these subcategories, and the one with the highest similarity score is chosen.

3. Deep-Level Matching

The final step involves matching the clinical text with the **specific codes** within the selected subcategory. This level provides the most granular match, narrowing the possibilities to a single ICD-10-CM code.

At each stage, the process uses a semantic similarity score to guide decision-making, ensuring that the clinical text aligns meaningfully with the selected code.

Semantic Similarity Computation

To calculate similarity, the methodology employs the `sentence_transformers` library. This library uses transformer-based models, such as BERT or RoBERTa, to generate embeddings for text. The embeddings capture the semantic essence of the input text and enable meaningful comparisons.

Cosine similarity is used as the metric to measure closeness between the embedding of the clinical text and that of ICD-10-CM descriptions. A similarity score closer to 1 indicates higher relevance.

Challenges and Limitations

Over-convergence to a Single Code

One of the significant issues with this approach is its tendency to converge to a single ICD-10-CM code at the deepest level of the hierarchy. Clinical documents often describe multiple conditions or symptoms in detail, which means they are naturally associated with more than one ICD-10-CM code. For example, a document discussing "hypertension and diabetes" should ideally map to codes for both conditions. The current methodology, by selecting only the most relevant code based on similarity scores, risks oversimplifying the clinical scenario.

Mismatch in Text Length and Detail

Clinical documents are typically lengthy paragraphs, filled with detailed descriptions of symptoms, diagnoses, and context. On the other hand, the descriptions of ICD-10-CM codes at each level of the hierarchy are usually concise, often one-line summaries. This inherent mismatch in text length and detail makes it difficult to compare the two effectively using semantic similarity models. Longer clinical documents may include unrelated or secondary information that dilutes the similarity score with shorter, more specific ICD descriptions.

Loss of Relevant Information

The hierarchical narrowing process introduces another limitation. At each level of the hierarchy, only the top match is selected for further exploration, discarding other potential matches. This leads to a loss of granularity and may exclude relevant codes early in the process. For instance, if a clinical document contains elements related to multiple top-level categories (e.g., both respiratory and cardiovascular conditions), the model may incorrectly focus on just one category, ignoring the others.

Conclusion

The current approach demonstrates a methodical and scalable solution for mapping clinical text to ICD-10-CM codes using semantic similarity. However, its over-convergence to a single code and lack of contextualization makes it less suitable for real-world scenarios, where clinical documents often relate to multiple codes.

Automating Medical Coding Using MedCAT and BERT

Introduction

Medical scribing, coding, and scrubbing are integral parts of healthcare documentation. **Scribing** involves recording interactions between clinicians and patients, **coding** converts diagnoses and procedures into standardized formats like ICD-10-CM codes, and **scrubbing** ensures these codes are accurate and compliant. Despite their importance, these processes are labor-intensive, requiring skilled coders to parse unstructured clinical text manually. This leads to delays, inconsistencies, and higher costs.

To address these inefficiencies, we aimed to automate medical coding, focusing on classifying clinical text by specialty, detecting relevant medical terms, and mapping these terms to ICD-10-CM codes. Initially, we experimented with a **semantic similarity-based approach** to achieve this goal. However, limitations of this method led us to adopt a more robust solution combining **MedCAT** and **BERT**. This documentation details the journey from our initial experiment to the implementation of the final system.

Initial Experiment: Semantic Similarity-Based ICD-10-CM Code Mapping

Objective

The semantic similarity approach aimed to assign ICD-10-CM codes to clinical text by leveraging transformer-based embeddings to calculate similarity between clinical documents and ICD-10-CM descriptions. By exploiting the hierarchical structure of ICD-10-CM codes, we progressively narrowed the

potential matches, starting with broad categories and moving to specific subcategories.

Challenges Encountered

While the methodology was conceptually sound, it presented several practical limitations:

1. **Over-Convergence to a Single Code:** The stepwise narrowing often resulted in assigning one code, ignoring that clinical text frequently involves multiple conditions.
2. **Mismatch in Text Granularity:** Clinical text is detailed and verbose, whereas ICD-10-CM descriptions are concise, leading to inconsistent similarity scores.
3. **Loss of Relevant Information:** The hierarchical narrowing discarded potential matches too early, reducing accuracy.

These limitations highlighted the need for a more comprehensive and context-aware solution, prompting our transition to **MedCAT** and fine-tuned **BERT**.

Enhanced Solution: Combining MedCAT and BERT

To overcome the limitations of the semantic similarity approach, we developed a two-stage system:

1. **Fine-tuned BERT** for medical specialty classification.
2. **MedCAT** for Named Entity Recognition and Linking (NER-L), followed by mapping terms to **ICD-10-CM codes**.

Step 1: Medical Specialty Classification Using BERT

Methodology

We fine-tuned a **BERT** model to classify clinical text by medical specialty. This step provides context, narrowing down the domain of the subsequent annotation process.

1. **Dataset:** A curated dataset of clinical text labeled by specialty (e.g., Cardiology, Endocrinology).
2. **Training:** The model was fine-tuned using a multi-class classification head.

3. **Evaluation:** Metrics such as accuracy, precision, recall, and F1-score were used to assess performance.

Outcome

The fine-tuned BERT model demonstrated good accuracy, reliably predicting the specialty of given clinical text and improving the downstream annotation process.

Step 2: Named Entity Recognition and Linking with MedCAT

Data Annotation Using MedCAT

We employed **CAT** model to annotate clinical text. The steps included:

- Extracting terms related to diseases (**T047**) and mental health conditions (**T048**).
- Linking these terms to their corresponding **Concept Unique Identifiers (CUIs)** using UMLS.

Fine-Tuning MedCAT

To enhance its performance, we fine-tuned MedCAT on a domain-specific dataset, improving its ability to recognize medical terms within the context of clinical text.

Mapping CUIs to ICD-10-CM Codes

Using the **UMLS Metathesaurus API**, the CUIs detected by MedCAT were mapped to their corresponding ICD-10-CM codes. This ensured a robust and scalable linkage between clinical terms and standardized codes.

Step 3: Streamlit Prototype for Demonstration

We built a **Streamlit** application as a prototype for the system. The app provided an intuitive interface for users to input unstructured clinical text and obtain structured insights, including:

1. **Medical Specialty:** Predicted using the fine-tuned BERT model.
2. **NER-L Results:** Detected medical terms with their respective CUIs.
3. **ICD-10-CM Codes:** Retrieved from UMLS for each detected CUI.

This prototype showcased the practicality and user-friendliness of our solution, emphasizing its potential for real-world deployment.

Comparing the Approaches

Feature	Semantic Similarity Approach	MedCAT and BERT Approach
Code Mapping	Single ICD-10-CM code, hierarchical	Multiple ICD-10-CM codes, comprehensive
Text Context	Limited to similarity embeddings	Enhanced through fine-tuned MedCAT
Accuracy	Limited due to hierarchical narrowing	High due to NER-L and BERT
Scalability	Moderate	High
Prototype Feasibility	Not implemented	Fully functional with Streamlit

The MedCAT and BERT approach resolved the shortcomings of the semantic similarity method, providing a robust, accurate, and scalable solution.

Results and Achievements

- Accurate Specialty Detection:** The fine-tuned BERT model reliably classified clinical text into medical specialties.
- Precise Annotation:** MedCAT annotated disease and mental health terms accurately, linking them to CUIs.

3. **Scalable Code Mapping:** The integration with UMLS provided robust mapping to ICD-10-CM codes.
4. **User-Friendly Interface:** The Streamlit prototype made the system accessible and practical for users.

Conclusion

Our project successfully automated medical coding by combining BERT for specialty classification and MedCAT for term annotation and ICD-10-CM mapping. While the initial semantic similarity approach had conceptual strengths, its limitations necessitated a shift to this enhanced methodology. The final system is scalable, accurate, and user-friendly, offering significant potential to transform medical coding in clinical workflows. With further enhancements, such as support for additional medical datasets and specialties, this system could address broader challenges in healthcare documentation.